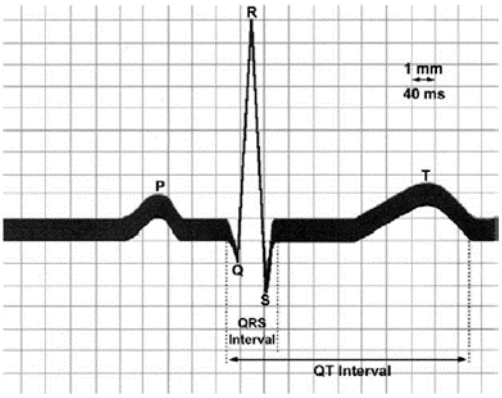
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Drug-induced QT Prolongation and Risk of TdP in Adult Patients



The QT interval encompasses ventricular depolarization (QRS) and ventricular repolarization (T). Delayed repolarization may precipitate torsades de pointes (TdP), which can result in the development of life-threatening ventricular fibrillation. Some drugs may prolong the QT interval and place patients at risk for TdP. Risk factors for drug-induced TdP include the following:

- Diagnosed/undiagnosed congenital long-QT syndrome (LQTS)
- Heart disease, especially congestive heart failure and myocardial infarction
- Electrolyte abnormalities (hypokalemia, hypomagnesemia, hypocalcemia)
- Bradycardia
- Drug-drug interactions resulting in increased exposure to drugs that prolong the QT interval
- Multiple QT-prolonging agents (and relative risk of each agent)
- Female sex

In the hospital, automated QT interpretations utilize the Bazett correction formula to adjust for cycle length. This resultant QTc is subject to several considerations, including:

- Over-corrects QT at faster heart rates (> 100 bpm) and under-corrects QT at lower heart rates (< 60 bpm), necessitating the use of alternative QT correction such as Fridericia formula. This corrected value is available in MiChart ECG reports.
- Manual read may be necessary for difficult-to-read ECGs or in those with irregular rhythms, such as atrial fibrillation
- Prolonged QRS > 120 ms (such as with left or right bundle branch block) may result in prolonged QT despite normal repolarization physiology. Rautaharju correction equation is preferred in this scenario. Corrected QTc for those with long QRS can be done using the Excel calculator provided on the Antimicrobial Stewardship Webpage

The Table below classifies agents by risk of significant QT prolongation, and thus, potential risk of TdP. The Table is not absolutely comprehensive. https://crediblemeds.org/ is a comprehensive resource (registration required). However, as this resource is comprehensive of all published reports, the following note is important:

Aside from agents in the RED category below, events of drug-induced TdP are extremely rare. Comprehensive avoidance of agents that may prolong the QT frequently confers greater risk of harms (decreased efficacy, other toxicities and/or interaction concerns, etc.) than the resultant reduction in risk of TdP.

The following are general recommendations regarding use of potentially QT-prolonging medications:

- Modifiable risk factors should be identified and remedied. Examples of this are:
    - Assess use of other QT-prolonging agents and query opportunities for de-prescribing and/or transition to more favorable therapies
    - Correct electrolyte abnormalities
- Initiation of an agent in the Red category requires pre-initiation (and sometimes post-initiation) ECG monitoring; see package inserts for detailed information.
- Initiation of an agent in the Orange category rarely requires pre- and post- initiation ECG assessment. Post-initiation assessment includes ECG assessment in the first 24 hours after initiation and after presumed steady-state.
    - ECG assessment *should* be undertaken prior to initiating an Orange agent in patients with known congenital LTQS or for patients currently receiving a Red agent. Initiation should be avoided if QTc > 500 or > 550 with ventricular conduction abnormalities or ventricular pacing. Post-initiation monitoring is recommended.
    - ECG monitoring may be *considered* prior to initiating an Orange agent in patients currently receiving other Orange agents and with multiple other risk factors for drug-induced TdP. Initiation should be avoided if QTc > 500 or > 550 with unadjusted ventricular conduction abnormalities or ventricular pacing. Post-initiation monitoring may also be considered.
- Greatest concern for drug-induced QT prolongation leading to TdP is QTc > 500 ms or > 550 ms with unadjusted intraventricular conduction delay (QRS > 120 ms) or ventricular pacing and/or > 60 ms increase in QTc from baseline
- In patients with concerning automated QTc reads, assess for limitations as described above.
- While the presence of an ICD may reduce the risk of death related to TdP, the same consideration for QT-prolongation should still be given to this population. For patients with ICDs (± LVADS) with refractory ventricular arrhythmias, the risk/benefit may be to tolerate a prolonged QTc.

Agents in which initiation of other QT-prolonging agents should be undertaken with extreme caution

Anti-arrythmics (Class IA and III)*:* Dofetilide, Disopyramide, Ibutilide, Procainamide, Quinidine, Sotalol

Anti-neoplastic agents: Arsenic trioxide, Ribociclib, Quizartinib, Vandetanib

Miscellaneous: Thioridazine

Agents with QT-prolonging effects that are almost always safe to use EXCEPT when multiple risk factors are present.

Macrolides: Erythromycin > Clarithromycin >> Azithromycin

Azole Antifungals: Fluconazole = Itraconazole = Voriconazole > Posaconazole

Fluoroquinolones: Moxifloxacin > Levofloxacin

Anti-Tuberculous: Bedaquiline, Clofazimine

Antipsychotics/Antidepressants *(Note: Unlikely after chronic use; typically dose-related; more likely in overdose):*
Droperidol (> 10 mg), Haloperidol (> 10 mg), Ziprasidone, Citalopram (≥ 40 mg), Tricyclic Antidepressants

Anti-neoplastic agents *(Note: for other agents, if the prescribing information indicates the drug causes QT prolongation please consult with Heme/Onc PharmD):*
Ivosidenib, Nilotinib

Miscellaneous: Alfuzosin, Chloroquine, Dronedarone, Flecainide, Indapamide, Methadone, Pentamidine (when given daily IV), Ranolazine, Vardenafil

Agents with either very mild or negligible effects on QT and are thus safe to use in patients (not all-inclusive, focus is on agents in classes mentioned above and/or commonly utilized agents that are often implicated as being at risk)

Azole Antifungals: Isavuconazole (significantly *decreases* QT interval)

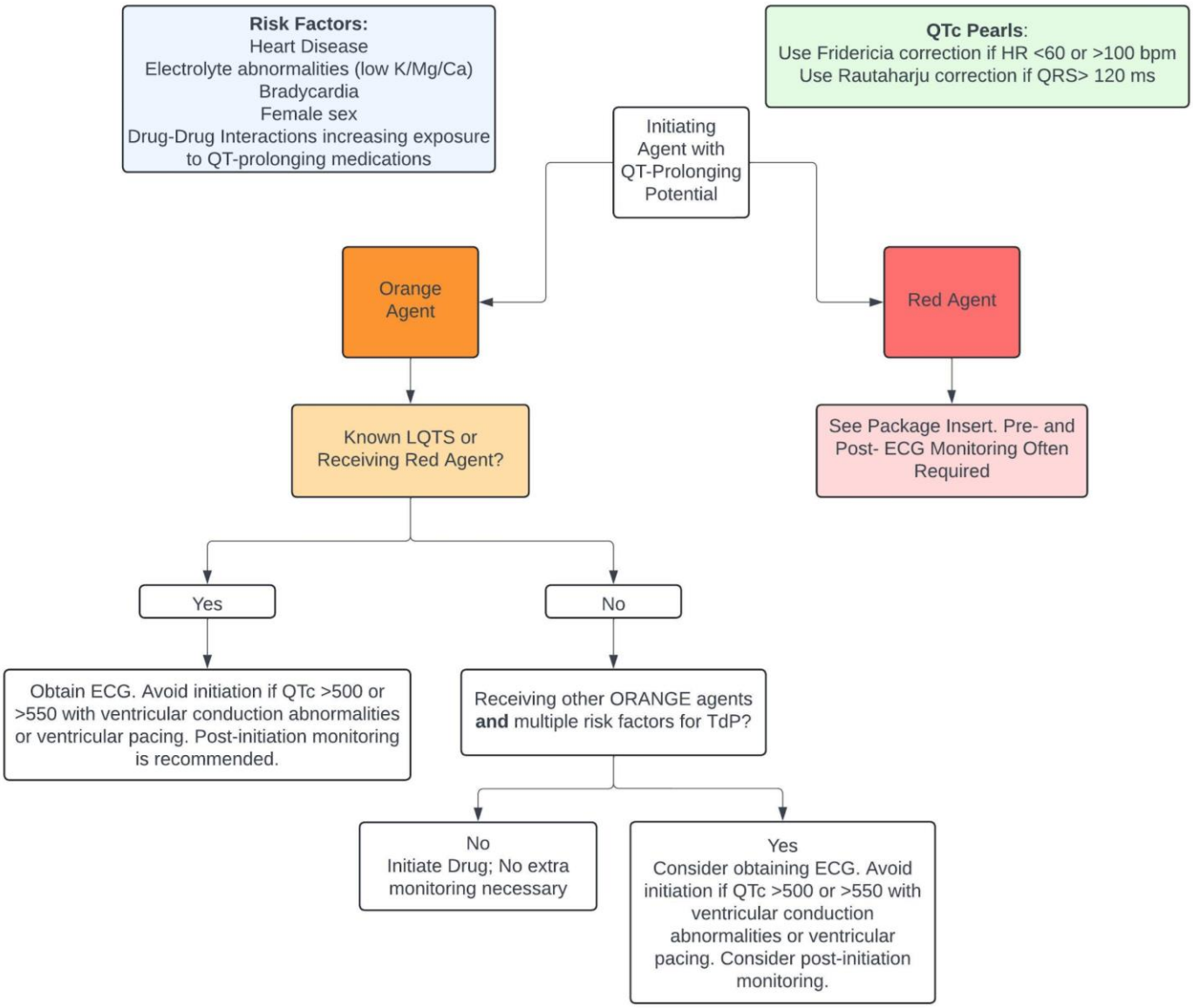
Anti-Emetics *(Note: QTc prolongation is minimal when used at clinical doses. QTc prolongation should not be an indication to use trimethobenzamide over these agents, as this agent is generally less effective for nausea and vomiting)*:
Granisetron, Olanzapine, Ondansetron, Palonosetron, Prochlorperazine

Fluoroquinolones: Ciprofloxacin

Amiodarone: *Note: Amiodarone, a class III anti-arrhythmic, markedly prolongs the QT but is rarely associated with TdP. There is extensive experience with safe use of agents with mild-moderate QT-prolonging effects (such as azole antifungals and fluoroquinolones) in patients receiving chronic amiodarone*.

Psychotropics: Other antidepressants and antipsychotics not listed above including Clozapine, Droperidol (≤ 10 mg), Haloperidol (≤ 10 mg), Olanzapine, Quetiapine, Risperidone

Figure:



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References

	Citation
1	Drew BJ et al. Prevention of Torsades de Pointes in Hospital Settings. JACC Vol. 55, No. 9, 2010.
2	https://www.fda.gov/regulatory-information/search-fda-guidance-documents/qtc-information-human-prescription-drug-and-biological-product-labeling
3	Owens RC, Nolin TD. Antimicrobial-Associated QT Interval Prolongation: Pointes of Interest. Clinical Infectious Diseases 2006; 43:1603–11.
4	Rautaharju PM et al. Assessment of Prolonged QT and JT Intervals in Ventricular Conduction Defects. Am J Cardiol 2004;93:1017–1021

Revision History

Date	**Revision**

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.